Trigonometric Identities — Problem Set

Warm-up

1. Use the compound angle formula for $\tan(A - B)$ to simplify $\frac{\tan 5\theta - \tan 2\theta}{1 + \tan 5\theta \tan 2\theta}$.
2. Expand $\sin(2x + x)$ using the sine sum formula.
3. If $\cos\theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$, find the exact value of $\sin 2\theta$.
4. State the three different forms of the double angle formula for $\cos 2A$.
5. Simplify the expression $2 \sin 15° \cos 15°$ using a double angle identity.
6. Given $\tan A = 2$, find the value of $\tan 2A$.
7. Rewrite $1 - 2\sin^2(3x)$ as a single trigonometric function.
8. Simplify $\frac{\sin 2\theta}{2 \sin\theta}$.
9. Express $\cos^2\alpha - \sin^2\alpha$ in terms of 2α.
10. Use the identity for $\sin(A - B)$ to find the exact value of $\sin 15°$ (take $A = 45°, B = 30°$).

Skill-building

1. Show that $\frac{\sin 2x}{1+\cos 2x} = \tan x$.
2. Prove the identity $\frac{\cos A - \cos 3A}{\sin 3A - \sin A} = \tan 2A$.
3. Solve $2 \sin\theta \cos\theta = \frac{1}{2}$ for $0 \leq \theta \leq \pi$.
4. Given $\cos 2\theta = \frac{7}{25}$ and $\pi < 2\theta < \frac{3\pi}{2}$, find the exact value of $\cos\theta$.
5. Prove that $\cos^4\theta - \sin^4\theta = \cos 2\theta$.
6. Simplify the expression $\frac{\sin 3\alpha}{\sin\alpha} - \frac{\cos 3\alpha}{\cos\alpha}$ to a constant integer.
7. Find all solutions to $\cos 2x + 3\cos x + 2 = 0$ in the domain $0 \leq x \leq 2\pi$.
8. Prove that $\csc 2\theta + \cot 2\theta = \cot\theta$.
9. If $\tan x = t$, show that $\sin 2x = \frac{2t}{1+t^2}$.
10. Use the product-to-sum identities to express $\sin 5x \cos 3x$ as a sum of two sine functions.

Easier Exam Questions

1. (Hornsby Girls 2021 Q5) Which expression is equivalent to $\frac{\tan 3x - \tan x}{1+\tan 3x \tan x}$?

   (A) $\tan 2x$

   (B) $\tan 4x$

   (C) $\tan 6x$

   (D) $\frac{\tan 3x - 1}{1+\tan 3x}$

2. (Girraween 2024 Q4)

   Which expression is equivalent to $\frac{1+\tan 5x \tan 3x}{\tan 5x - \tan 3x}$

   A. $\cot 2x$

   B. $\tan 2x$

   C. $\frac{\tan 2x}{1+\tan 5x \tan 3x}$

   D. $\cot x$

3. (Baulkham Hills 2024 Q5)

   Which of the following is equivalent to $\frac{1-\cos 2x}{\sin 2x}$?

   A. $1 - \cot 2x$

   B. 1

   C. $\cot x$

   D. $\tan x$

4. (Hornsby Girls 2021 Q4) The angle θ satisfies $\sin\theta = \frac{12}{13}$ for $\frac{\pi}{2} < \theta < \pi$.

   The value of $\cos 2\theta$ is...

   (A) $\frac{10}{13}$

   (B) $-\frac{10}{13}$

   (C) $\frac{119}{169}$

   (D) $-\frac{119}{169}$

5. (Presbyterian 2023 Q12b)

   Simplify $\frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x}$ **2**

6. (Gosford 2020 Q9c)

   Show that $\frac{1-\cos 2x + \sin 2x}{1+\cos 2x + \sin 2x} = \tan x$. **2**

Harder Exam Questions

1. (Blacktown Boys 2022 Q22b)

   $2\cos^2 2x - \cos 2x - 1 = 0,\ 0 \le x \le \pi$ **3**

2. (Blacktown Boys 2022 Q24)

   Prove that $\tan\theta(1+\tan\theta) + (1+\cot\theta) = \frac{\tan\theta + 1}{\sin\theta\cos\theta}$ **3**

3. (Girraween 2024 Q4 a, b, d & e)

(a) Using the expansion of $\cos(A+B)$, find the exact value of $\sec 105°$ **3**

(b)

i. Prove that $\tan\theta = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}}$ **2**

ii. Hence show the exact value of $\tan 22\frac{1}{2}^{\circ} = \sqrt{2}-1$ **2**

(d) Show that $\sin(A-2B)\cos(A+2B) = \frac{1}{2}(\sin 2A - \sin 4B)$ **2** Note: Sine is an odd function and Cosine is an even function.

(e) Prove the identity $\frac{2\sin^3 A + 2\cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$ **2**

4. (Baulkham Hills 2024 Q11c)

(i) Show that $\cot\theta - \cot 2\theta = \text{cosec } 2\theta$. **2**

(ii) Hence find the exact value of $\cot 15°$. **1**

5. (Sydney Girls 2021 Q4d)

Prove that: $\tan\left(\frac{\pi}{4}+\theta\right) + \tan\left(\frac{\pi}{4}-\theta\right) = 2\sec 2\theta$ **3**

6. (Sydney Girls 2020 Q4c)

Suppose that x is a real number and that m and n are positive integers.

(i) Express $\sin mx \sin\frac{x}{2}$ as a difference of cosine functions. **1**

(ii) Hence, show that **2**

$$\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\cos\frac{x}{2} - \cos\left(n+\frac{1}{2}\right)x}{2\sin\frac{x}{2}}.$$

7. (Sydney Girls 2022 Q2b)

(i) Prove that $(\sin\alpha - \cos\alpha)^2 = 1 - \sin 2\alpha$. **2**

(ii) Hence, or otherwise, find the exact value of $\sin 75° - \cos 75°$. Leave your answer with a rational denominator. **2**

8. (Normanhurst Boys 2019 Q15 a,b& c)

(a) Find the exact value of $(\sin 22.5° + \cos 22.5°)^2$ **2**

(b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$. **4**

Find the exact value of $\tan(x+y)$.

(c) Prove that $\frac{\sin 3\theta}{\sin\theta} + \frac{\cos 3\theta}{\cos\theta} = 4\cos 2\theta$ **3**

9. (Caringbah 2025 Q8b)

(a) Find the exact value of $(\sin 22.5° + \cos 22.5°)^2$ **2**

(b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$. **4**

Find the exact value of $\tan(x+y)$.

(c) Prove that $\frac{\sin 3\theta}{\sin\theta} + \frac{\cos 3\theta}{\cos\theta} = 4\cos 2\theta$ **3**

10. (Caringbah 2025 Q10c)

Solve **4**

$$\cos^2 x + \cos^2(2x) + \cos^2(3x) = 1, 0 \leq x \leq 2\pi$$